

heat dissipation than a shorter and thinner one. However, the former might not be easily compatible with your case and you might need to make some adjustments for the dimensions. For example, you could go with a longer but thinner radiator. Another thing to remember is that the longer your radiator, the greater number of fans you need to install. And as the number of fans increases, the acoustics grow louder. This is also the case with a thicker radiator since you will need to install a fan with a high static pressure that will lead to louder acoustics.

The final consideration should be the fin density measured in FPI or fins per inch. So, when you scroll down to the specifications of the radiator, you will come across the FPI field and find a value around 20 or 30. What you need to remember is higher the FPI, better the dissipation of heat. However, as the FPI increases, the fins become denser. This makes it difficult for airflow and you will have to opt for high static pressure fans that will be loud.

We've already mentioned how the choice of a fan can affect your radiator preferences. If you're sensitive to noise, then you should avoid thick and high FPI radiators. Of course, the size of the fans, whether 120mm or 140mm, and the number of fans will definitely help you decide the length of the radiator. While you're



Denser fins will provide better dissipation of heat but louder acoustics

picking the radiator thickness, you also need to consider whether the case will accommodate the fans as well and not just the radiator. The maximum RPM needs to be considered as well since that will determine how fast the fan rotates and how loud it could get. Higher the RPM, louder the fan. You should be comfortable with a fan at around 1000 RPM. Anything higher than that, and you will start noticing the sound coming from the airflow. Based on the thickness of the radiator and FPI, you should choose a low, medium or high static pressure fan. You don't want to end up with a fan that doesn't cool down the radiator and hence, the fluid. Also, there are options out there for RGB lighting enabled radiator fans that can be used to match or